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`graphicx`
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Inverse Kinematic Solution for EE543_G6

Friday 21st March, 2025

1 Introduction

This report describes closed form inverse kinematics solutions for EE543_G6. The solution was generated by the IK-BT package from the University of Washington Biorobotics Lab. The IK-BT package is described in <https://arxiv.org/abs/1711.05412>. IK-BT derives your equations using Python 3.8 and the sympy 1.9 module for symbolic mathematics.

2 Kinematic Parameters

The kinematic parameters for this robot are

$$[\alpha_{i-1}, \quad a_{i-1}, \quad d_i, \quad \theta_i]$$

$$\begin{bmatrix} 0.0 & 0.0 & l_1 & th_1 \\ \frac{\pi}{2} & 0.0 & 0.0 & th_2 \\ 0.0 & a_3 & 0.0 & th_3 \\ \frac{\pi}{2} & 0.0 & l_2 & th_4 \\ 0.0 & 0.0 & d_1 & 0.0 \\ 0.0 & 0.0 & 0.0 & 0.0 \end{bmatrix} \quad (1)$$

3 Forward Kinematic Equations

The forward kinematic equations for this robot are:

$$\begin{array}{lcl}
 \begin{bmatrix} r_{11} & r_{12} & r_{13} & Px \\ r_{21} & r_{22} & r_{23} & Py \\ r_{31} & r_{32} & r_{33} & Pz \\ 0 & 0 & 0 & 1 \end{bmatrix} & & \\
 \text{Column } 1 & \begin{bmatrix} c_4 (c_1 c_2 c_3 - c_1 s_2 s_3) + s_1 s_4 \\ -c_1 s_4 + c_4 (c_2 c_3 s_1 - s_1 s_2 s_3) \\ c_4 (c_2 s_3 + c_3 s_2) \\ 0 \end{bmatrix} & \\
 \text{Column } 2 & \begin{bmatrix} c_4 s_1 - s_4 (c_1 c_2 c_3 - c_1 s_2 s_3) \\ -c_1 c_4 - s_4 (c_2 c_3 s_1 - s_1 s_2 s_3) \\ -s_4 (c_2 s_3 + c_3 s_2) \\ 0 \end{bmatrix} & (2) \\
 \text{Column } 3 & \begin{bmatrix} c_1 c_2 s_3 + c_1 c_3 s_2 \\ c_2 s_1 s_3 + c_3 s_1 s_2 \\ -c_2 c_3 + s_2 s_3 \\ 0 \end{bmatrix} & \\
 \text{Column } 4 & \begin{bmatrix} a_3 c_1 c_2 + d_1 (c_1 c_2 s_3 + c_1 c_3 s_2) - l_2 (-c_1 c_2 s_3 - c_1 c_3 s_2) \\ a_3 c_2 s_1 + d_1 (c_2 s_1 s_3 + c_3 s_1 s_2) - l_2 (-c_2 s_1 s_3 - c_3 s_1 s_2) \\ a_3 s_2 + d_1 (-c_2 c_3 + s_2 s_3) + l_1 - l_2 (c_2 c_3 - s_2 s_3) \\ 1 \end{bmatrix} &
 \end{array}$$

Note: column numbers use math notation rather than python indeces.

4 Unknown Variables:

The unknown variables for this robot are (in solution order):

1. θ_1
2. θ_2
3. θ_3
4. θ_4
5. θ_{23}

5 Solutions in Generic Form

The following equations comprise solutions for each unknown.

5.1 θ_1

Solution Method: $\text{atan2}(y,x)$

$$\theta_{1s1} = \text{atan2}(r_{23}, r_{13}) \quad (3)$$

$$\theta_{1s2} = \text{atan2}(-r_{23}, -r_{13}) \quad (4)$$

5.2 θ_2

Solution Method: $\arcsin$

$$\theta_{2s1} = \arcsin\left(\frac{Pz - d_1 r_{33} - l_1 - l_2 r_{33}}{a_3}\right) \quad (5)$$

$$\theta_{2s2} = \pi - \arcsin\left(\frac{Pz - d_1 r_{33} - l_1 - l_2 r_{33}}{a_3}\right) \quad (6)$$

5.3 θ_3

Solution Method: atan2(y,x), arcsin, best ranked, atan2(y,x)

$$\begin{aligned} \theta_{3s1} &= \text{atan}_2(r_{13} \cos(\theta_1) \cos(\theta_2) + r_{23} \sin(\theta_1) \cos(\theta_2) + r_{33} \sin(\theta_2), r_{13} \sin(\theta_2) \cos(\theta_1) + r_{23} \sin(\theta_1) \sin(\theta_2) - r_{33} \cos(\theta_2)) \end{aligned} \quad (7)$$

5.4 θ_4

Solution Method: atan2(y,x), arcsin, best ranked, atan2(y,x)

$$\theta_{4s1} = \text{atan}_2(r_{11} \sin(\theta_1) - r_{21} \cos(\theta_1), r_{12} \sin(\theta_1) - r_{22} \cos(\theta_1)) \quad (8)$$

5.5 θ_{23}

Solution Method: algebra

$$\theta_{23s1} = \theta_2 + \theta_3 \quad (9)$$

6 Solutions to Generate all Versions

The following equations are the full set of solutions for each unknown incorporating all combinations of dependencies.

6.1 θ_1

Solution Method: atan2(y,x)

$$\theta_{1v1} = \text{atan}_2(r_{23}, r_{13}) \quad (10)$$

$$\theta_{1v2} = \text{atan}_2(r_{23}, r_{13}) \quad (11)$$

$$\theta_{1v3} = \text{atan}_2(r_{23}, r_{13}) \quad (12)$$

$$\theta_{1v4} = \text{atan}_2(r_{23}, r_{13}) \quad (13)$$

6.2 θ_2

Solution Method: arcsin

$$\theta_{2v1} = \text{asin}\left(\frac{Pz - d_1 r_{33} - l_1 - l_2 r_{33}}{a_3}\right) \quad (14)$$

$$\theta_{2v2} = \text{asin}\left(\frac{Pz - d_1 r_{33} - l_1 - l_2 r_{33}}{a_3}\right) \quad (15)$$

$$\theta_{2v3} = \text{asin}\left(\frac{Pz - d_1 r_{33} - l_1 - l_2 r_{33}}{a_3}\right) \quad (16)$$

$$\theta_{2v4} = \text{asin}\left(\frac{Pz - d_1 r_{33} - l_1 - l_2 r_{33}}{a_3}\right) \quad (17)$$

6.3 θ_3

Solution Method: atan2(y,x), arcsin, best ranked, atan2(y,x)

$$\theta_{3v1} = \text{atan}_2(r_{13} \cos(\theta_{1s1}) \cos(\theta_{2v1}) + r_{23} \sin(\theta_{1s1}) \cos(\theta_{2v1}) + r_{33} \sin(\theta_{2v1}), r_{13} \sin(\theta_{2v1}) \cos(\theta_{1s1}) + r_{23} \sin(\theta_{1s1}) \sin(\theta_{2v1}) - r_{33} \cos(\theta_{2v1})) \quad (18)$$

$$\theta_{3v2} = \text{atan}_2(r_{13} \cos(\theta_{1s2}) \cos(\theta_{2v2}) + r_{23} \sin(\theta_{1s2}) \cos(\theta_{2v2}) + r_{33} \sin(\theta_{2v2}), r_{13} \sin(\theta_{2v2}) \cos(\theta_{1s2}) + r_{23} \sin(\theta_{1s2}) \sin(\theta_{2v2}) - r_{33} \cos(\theta_{2v2})) \quad (19)$$

$$\theta_{3v3} = \text{atan}_2(r_{13} \cos(\theta_{1s2}) \cos(\theta_{2v3}) + r_{23} \sin(\theta_{1s2}) \cos(\theta_{2v3}) + r_{33} \sin(\theta_{2v3}), r_{13} \sin(\theta_{2v3}) \cos(\theta_{1s2}) + r_{23} \sin(\theta_{1s2}) \sin(\theta_{2v3}) - r_{33} \cos(\theta_{2v3})) \quad (20)$$

$$\theta_{3v4} = \text{atan}_2(r_{13} \cos(\theta_{1s1}) \cos(\theta_{2v4}) + r_{23} \sin(\theta_{1s1}) \cos(\theta_{2v4}) + r_{33} \sin(\theta_{2v4}), r_{13} \sin(\theta_{2v4}) \cos(\theta_{1s1}) + r_{23} \sin(\theta_{1s1}) \sin(\theta_{2v4}) - r_{33} \cos(\theta_{2v4})) \quad (21)$$

6.4 θ_4

Solution Method: atan2(y,x), arcsin, best ranked, atan2(y,x)

$$\theta_{4v1} = \text{atan2}(r_{11} \sin(\theta_{1s1}) - r_{21} \cos(\theta_{1s1}), r_{12} \sin(\theta_{1s1}) - r_{22} \cos(\theta_{1s1})) \quad (22)$$

$$\theta_{4v2} = \text{atan2}(r_{11} \sin(\theta_{1s2}) - r_{21} \cos(\theta_{1s2}), r_{12} \sin(\theta_{1s2}) - r_{22} \cos(\theta_{1s2})) \quad (23)$$

$$\theta_{4v3} = \text{atan2}(r_{11} \sin(\theta_{1s2}) - r_{21} \cos(\theta_{1s2}), r_{12} \sin(\theta_{1s2}) - r_{22} \cos(\theta_{1s2})) \quad (24)$$

$$\theta_{4v4} = \text{atan2}(r_{11} \sin(\theta_{1s1}) - r_{21} \cos(\theta_{1s1}), r_{12} \sin(\theta_{1s1}) - r_{22} \cos(\theta_{1s1})) \quad (25)$$

6.5 θ_{23}

Solution Method: algebra

$$\theta_{23v1} = \theta_{2v1} + \theta_{3v1} \quad (26)$$

$$\theta_{23v2} = \theta_{2v2} + \theta_{3v2} \quad (27)$$

$$\theta_{23v3} = \theta_{2v3} + \theta_{3v3} \quad (28)$$

$$\theta_{23v4} = \theta_{2v4} + \theta_{3v4} \quad (29)$$

7 Solution Graph (Edges)

The following is the abstract representation of solution graph for this manipulator (nodes with parent -1 are roots). Future: graphic representation. :

Edge:th_23 depends on: th_3 (th_23-->th_3) Edge:th_3 depends on: th_1 (th_3-->th_1)
 Edge:th_23 depends on: th_2 (th_23-->th_2) Edge:th_4 depends on: th_1 (th_4-->th_1)
 Edge:th_3 depends on: th_2 (th_3-->th_2)

8 Solution Set

The following are the sets of joint solutions (poses) for this manipulator:

('th_1s2', 'th_2v3', 'th_3v3', 'th_4v3', 'th_23v3')
 ('th_1s1', 'th_2v1', 'th_3v1', 'th_4v1', 'th_23v1')
 ('th_1s1', 'th_2v4', 'th_3v4', 'th_4v4', 'th_23v4')
 ('th_1s2', 'th_2v2', 'th_3v2', 'th_4v2', 'th_23v2')

9 Solution Set v3

th_1s2	th_2v3	th_3v3	th_4v3	th_23v3
th_1s1	th_2v1	th_3v1	th_4v1	th_23v1
th_1s1	th_2v4	th_3v4	th_4v4	th_23v4
th_1s2	th_2v2	th_3v2	th_4v2	th_23v2

10 Jacobian Matrix

$$\begin{aligned}
 {}^6J_6 = & \\
 & \text{Column 1} \\
 & \begin{bmatrix} -a_3c_2s_4 - c_2d_1s_3s_4 - c_2l_2s_3s_4 - c_3d_1s_2s_4 - c_3l_2s_2s_4 \\ -a_3c_2c_4 - c_2c_4d_1s_3 - c_2c_4l_2s_3 - c_3c_4d_1s_2 - c_3c_4l_2s_2 \\ 0 \\ c_2c_4s_3 + c_3c_4s_2 \\ -c_2s_3s_4 - c_3s_2s_4 \\ -c_2c_3 + s_2s_3 \end{bmatrix} \\
 & \text{Column 2} \\
 & \begin{bmatrix} a_3c_4s_3 + c_4d_1 + c_4l_2 \\ -a_3s_3s_4 - d_1s_4 - l_2s_4 \\ -a_3c_3 \\ s_4 \\ c_4 \\ 0 \end{bmatrix} \\
 & \text{Column 3} \\
 & \begin{bmatrix} c_4d_1 + c_4l_2 \\ -d_1s_4 - l_2s_4 \\ 0 \\ s_4 \\ c_4 \\ 0 \end{bmatrix} \\
 & \text{Column 4} \\
 & \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \\
 & \text{Column 5} \\
 & \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \\
 & \text{Column 6} \\
 & \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}
 \end{aligned} \tag{30}$$